Beyond Constant Fault Rates: Reliability Bounds for Gated Computation and Semantic Mechanization

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Abstract

Constant fault intensity, independent segments, fresh retries, and a fixed fault distribution are different assumptions. Earlier papers on gated computation use different subsets of them. We separate these assumptions and examine which conclusions survive their removal. A conditional survival identity replaces the independent-segment product. Uniform conditional gate accuracy and progress bounds preserve logarithmic verification overhead without a Poisson arrival model. A verification floor yields a finite horizon only when its exposure persists along correct prefixes. Unequal difficulty, latent regimes, persistent retry defects, and changing input distributions give counterexamples to predictions based on one average rate. Two positive results locate the $1/\sigma$ ceiling multiplier of the first paper. For the ideal gated survival curve, a latent rate drawn once per run preserves it exactly, and a monotone hazard orders it: at most $1/\sigma$ for an increasing failure rate and at least $1/\sigma$ for a decreasing one. With finite segments, the upper bounds hold after rounding to whole segments. For semantic mechanization, local escape and fallback probabilities replace one volume-weighted closure fraction. The twin result survives changes in the arrival law and dependent or adaptive sampling, provided twins induce the same observation law. We classify all 25 named results of the four earlier papers as arrival-free, conditionally preserved, or stationary-parameter formulas that need replacement, and we give executable finite-model checks. These are mathematical models, not measurements of deployed agents. The contribution is their application and synthesis; the probability identities and inequalities are classical.

1 Introduction

Part I models faults by a homogeneous Poisson process and derives a verification threshold, a stealth-limited horizon, and a checkpoint interval [1]. Part II studies representation blind spots [2]. Part III studies accepted-output escape, cost, and shared provenance [3]. Part IV studies progressive semantic mechanization (PSM): moving work from executor calls to deterministic mechanisms [4].

A natural question is whether their conclusions depend on faults entering at a constant rate. The answer differs by result. A blind-set inclusion is a statement about what a gate can distinguish. A horizon is a statement about how often unresolved faults occur over a sequence. One does not determine the other. Even a constant expected count per unit work need not give an exponential probability of completing without fault.

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Ord's analysis motivates the constant-hazard baseline on a particular task suite and leaves its wider validity open [5]. A fit of success against task duration also does not, by itself, identify within-run fault arrivals, retry behavior, or checkpoint completeness. We do not select an empirically correct alternative here. We identify the assumptions needed to use each result and provide countermodels when those assumptions are absent.

The contributions are: (i) a catalog of candidate fault-introduction models, mapped to the results they affect; (ii) conditional versions of the threshold and floor arguments; (iii) a cumulative-exposure extension, two results on when the $1/\sigma$ ceiling multiplier survives, and counterexamples to average-rate substitution; (iv) a separation of PSM's closure, evidence, drift, and cost assumptions; and (v) an assessment and classification of every named theorem, proposition, lemma, and corollary in Parts I–IV, with a list of errata candidates found on the way. We call a result *preserved* only with its remaining hypotheses. A failed extension does not refute a theorem inside its original model. Separate wording issues are identified as such.

2 What is allowed to change?

We distinguish six axes.

1. **Difficulty:** a segment's prior fault probability can depend on its work, length, and context load. A deterministic rate $\lambda(t)$ is one model.
2. **Dependence:** segments can share a latent regime, input, mechanism, or upstream fault. Constant marginal probabilities do not remove this dependence.
3. **Retry:** a rejected attempt can change the next attempt's distribution. A rollback can restore an artifact while retaining a faulty premise or environment.
4. **Fault type and visibility:** the distribution of produced faults and the gates' blind sets can change together. Fault intensity alone omits this interaction.
5. **Intervention:** repair, reset, routing, and mechanization can change later state. Their effect is not generally a substitution into a fixed sequence of risks.
6. **Semantics and workload:** input distributions, specifications, and mechanisms can drift. A deterministic mechanism can then have time-varying error without random execution.

Three clocks must also stay separate: useful work or emitted units, attempted work including retries, and calendar time. The first indexes reliability below. The second determines execution cost. The third matters for drift and maintenance. A median horizon on one clock is not a median on another. A fixed semantic defect is not a new arrival on every affected output.

Candidate fault-introduction models. Table 1 lists candidate replacements for the constant-rate model. Each is an established modeling family; none is selected here as empirically correct. The last column names the earlier results most sensitive to each model; Section 9 gives the full mapping. Adversarial fault introduction is not a stochastic process and is out of scope, as it is for Part I's benign-executor analysis.

Definition 1 (Success and emission)**.** Fix a workload and policy. Unless stated otherwise, each requested unit eventually emits with probability one. Let I_i indicate that emitted unit i has an escaped fault. Put $A_n = \{I_1 = \cdots = I_n = 0\}$, $S_n = \mathbb{P}(A_n)$, and $A_0 = \Omega$. Success means no escaped fault in the first n emitted units. Later repair does not erase an earlier escape in this metric. The discrete half-success horizon is $H_{1/2} = \inf\{n : S_n \le 1/2\}$, with $\inf \varnothing = \infty$.

Table 1: Candidate fault-introduction models and where this paper treats them.

Model	What changes	Treated in	Most sensitive earlier results
Nonhomogeneous Poisson, deterministic $\lambda(t)$, incl. Weibull [12]	difficulty by position	Props. 9, 11	ceiling multiplier, checkpoint interval, R-A
Mixed Poisson / frailty: rate drawn once per run [11]	dependence via a latent rate	Ex. 12, Prop. 13	horizon value; the multiplier only at finite s (ideal limit: exact)
Markov-modulated / regime switching [14]	dependence, persistence	Eq. (7)	slip, threshold uniformity
Self-exciting / propagation [9]	faults beget faults	Sec. 6, propagation	after an escape: damage only; before an escape: slip, horizon, retry cost
Common-cause / shared defect	one fault in many outputs	Ex. 3	series identity, horizon
Persistent retry / imperfect repair [13]	retry distribution	Prop. 5, Ex. 6, Eq. (7)	slip, checkpoint cost, R-C
Load- or context-dependent hazard	hazard as a function of state	Prop. 11; R-E row	R-E, ceiling multiplier
Visibility that co-varies with intensity	fault type × gate	Sec. 5, Prop. 13	diversity, ceiling
Input or specification drift	the fault set itself	Prop. 19	replay, evidence, validation

If emission can stall, report its probability and cost separately, or count non-completion as failure. Conditioning on eventual completion can select easier runs. We do not use that conditioning to conceal failures. Damage, the number of faulty outputs, and terminal correctness after repair are different metrics. They require additional models.

3 Conditional risk replaces constant rate

Proposition 2 (Survival and exposure)**.** *For each i with $S_{i-1} > 0$, define $q_i = \mathbb{P}(I_i = 1 \mid A_{i-1})$. Then*

$$S_n = \prod_{i=1}^{n}(1 - q_i), \qquad -\log S_n = \sum_{i=1}^{n} -\log(1 - q_i). \tag{1}$$

The second equality uses $-\log 0 = \infty$. After survival reaches zero, set later $q_i = 0$. If $\ell_i \le q_i \le u_i$ at every index where q_i is defined, with $0 \le \ell_i \le u_i \le 1$, then

$$\prod_i (1 - u_i) \le S_n \le \prod_i (1 - \ell_i), \qquad 1 - S_n \le \sum_i q_i \le \sum_i u_i.$$

If all $q_i < 1$, then $S_\infty > 0$ if and only if $\sum_i q_i < \infty$.

Proof. The chain rule gives $S_i = S_{i-1}(1 - q_i)$. Iterate and take logarithms. The product bounds follow factor by factor. The inequality $1 - \prod_i (1 - q_i) \le \sum_i q_i$ follows by induction. For the last claim, $-\log(1 - x) \ge x$. Conversely, summability makes $q_i \to 0$; eventually $q_i \le 1/2$, where $-\log(1 - q_i) \le 2q_i$. The finite prefix has positive product. □

The identity is the chain rule, not a new independence theorem. Its useful consequence is where bounds must hold: on histories that have remained correct. Unconditional marginals $f_i = \mathbb{P}(I_i = 1)$

cannot generally replace q_i. For arbitrary dependence they imply only

$$\max_i f_i \le \mathbb{P}\Big(\bigcup_i \{I_i = 1\}\Big) \le \min(1, \sum_i f_i). \tag{2}$$

These bounds are sharp: nested events attain the lower bound; disjoint events attain the upper bound when the sum is at most one. No homogeneous arrival process is needed for either bound.

Counterexample 3 (Same marginal risk, different horizons)**.** Let $f = 0.01$. Independent indicators give $S_{100} = 0.99^{100} \approx 0.3660$. If $I_i = Z$ for all i, with Z Bernoulli(0.01), then $S_n = 0.99$ for every $n \ge 1$. Here $q_1 = 0.01$ and $q_i = 0$ for $i > 1$. Both models have expected fault count $0.01n$. The second has no finite half-success horizon, but has probability 0.01 of all n outputs being faulty. Dependence changes the distribution of damage, not just a horizon constant.

Counterexample 4 (A floor that declines with exposure)**.** Let independent segment slips have $q_i = 0.1\,2^{-i}$. Every segment has positive risk, yet $\mathbb{P}(\text{any escape ever}) \le \sum_{i\ge1} q_i = 0.1$. Thus a positive local floor at each step does not imply an eventual escape with probability one, or a finite median horizon. A persistent lower bound, or sufficient cumulative exposure, is needed.

4 Retries and a threshold without Poisson arrivals

Proposition 5 (First accepted attempt)**.** *For a single unit, let R_r mean attempt r is reached, with $R_1 = \Omega$. Let $a_r = \mathbb{P}(\text{accept at } r \mid R_r)$ and $b_r = \mathbb{P}(\text{bad and accept at } r \mid R_r)$. If emission occurs almost surely, then*

$$\mathbb{P}(\text{emitted bad}) = \sum_{r\ge1} \mathbb{P}(R_r) b_r, \quad \mathbb{E}N = \sum_{r\ge1} \mathbb{P}(R_r), \quad \mathbb{P}(R_{r+1}) = \mathbb{P}(R_r)(1 - a_r). \tag{3}$$

If $a_r = a > 0$ and $b_r = b$ for every reached attempt, these reduce to b/a and $1/a$. More generally, if $a_r \ge a_0 > 0$ and $b_r \le b_0$, then $\mathbb{E}N \le 1/a_0$ and $\mathbb{P}(\text{emitted bad}) \le b_0/a_0$. A ratio bound is often sharper: if emission occurs almost surely and $b_r \le \kappa a_r$ at every reached attempt, then $\mathbb{P}(\text{emitted bad}) \le \min(1, \kappa)$.

Proof. First acceptance events are disjoint. Sum their bad parts. The tail-sum identity gives $\mathbb{E}N$. Conditional rejection gives the recursion. A geometric tail follows from $a_r \ge a_0$. Summing that tail proves both bounds and also establishes almost-sure emission in this case. For the ratio bound, $\sum_r \mathbb{P}(R_r) b_r \le \kappa \sum_r \mathbb{P}(R_r) a_r = \kappa$, because the last sum is the emission probability. □

The same reasoning applies conditional on a pre-unit history. Independent and identically distributed retries are sufficient for constant a, b, but are not necessary. In the usual notation, $a = (1-p)(1-\beta_m) + p\alpha_m$ and $b = p\alpha_m$.

Counterexample 6 (Persistent difficulty invalidates pooled retry arithmetic)**.** Choose a unit type once. With probability 1/2, it is always good and accepted. Otherwise it is always bad and each retry accepts independently with probability 1/2. Every unit eventually emits. A fresh, pooled first attempt has $a = 3/4$, $b = 1/4$, but the eventual bad fraction is 1/2, not $b/a = 1/3$. Expected attempts are $\frac{1}{2}(1) + \frac{1}{2}(2) = 3/2$, not $1/a = 4/3$. If bad units instead always reject, half the units never emit. A fresh first-attempt acceptance probability of 1/2 does not imply finite expected completion time.

Theorem 7 (Uniform conditional threshold)**.** *Fix n units and $0 < \delta < 1$. At every reachable pre-attempt history, including previous rejections and earlier slips, suppose:*

(i) the candidate is bad with probability at most $\bar{p} < 1$;

(ii) conditional on its good/bad status and that history, the m gate-error indicators are independent, each with probability at most $1/2 - \gamma$, for fixed $\gamma > 0$;

(iii) a majority decides acceptance; an attempt costs at most $s + mg$ for fixed s, g.

Choose odd m and write $d_m = e^{-2m\gamma^2}$. If

$$d_m \le \min\{1/2,\ \delta(1-\bar{p})/(2n\bar{p})\},$$

where the second constraint is omitted when $\bar{p} = 0$, then

$$S_n \ge 1 - \delta, \qquad \mathbb{E}[\textit{total cost}] \le \frac{2n(s+mg)}{1-\bar{p}}.$$

For fixed $\bar{p}, \gamma, s, g$ this gives $m = O(\log(n/\delta))$ and logarithmic multiplicative overhead. Neither Poisson arrivals nor independence across units is required.

Proof. Hoeffding's inequality bounds each majority error by d_m [6]. At every history, acceptance has probability at least $(1-\bar{p})(1-d_m) \ge (1-\bar{p})/2$, and bad acceptance has probability at most $\bar{p}d_m$. These uniform bounds remain valid after averaging over rejected histories. The ratio bound of Proposition 5, with $\kappa = 2\bar{p}d_m/(1-\bar{p})$, bounds each unit's slip conditional on every pre-unit history by $\kappa \le \delta/n$. Its geometric tail bounds the expected attempts by $2/(1-\bar{p})$. Average the slip bound over correct prefixes and apply Proposition 2. Sum conditional expected costs over all units, including histories with earlier slips. □

This theorem permits changing priors and adaptive history. Its uniform hypotheses are strong. A latent hard state can violate (i); shared gate blindness can violate (ii). Bounds only on correct prefixes suffice for the reliability calculation, but not for unconditional expected cost after a slip. A decreasing gate edge also changes the conclusion: the sufficient gate count scales as $\gamma^{-2}\log(n/\delta)$, not uniformly as $\log n$ if γ tends to zero.

Proposition 8 (Conditional verification floor)**.** *At each correct pre-unit history h, suppose the first attempt is bad with probability $p_i(h)$. Conditional on badness, a fraction $\sigma_i(h)$ passes every gate and the aggregator accepts unanimous passes. Then*

$$q_i \ge \mathbb{E}[p_i(h)\sigma_i(h) \mid A_{i-1}].$$

In particular, if $p_i(h)\sigma_i(h) \ge \ell_i$ uniformly, then $S_n \le \prod_i(1-\ell_i) \le \exp(-\sum_i \ell_i)$. If $\ell_i \ge \ell > 0$, then $H_{1/2} \le \lceil \log 2/[-\log(1-\ell)] \rceil$ for $\ell < 1$.

Proof. A stealth fault on the first attempt immediately emits a bad unit. Subsequent attempts can only add to this event, not remove it. Average over correct histories and use Proposition 2 and $1 - x \le e^{-x}$. □

Fresh retries are unnecessary for this lower bound. Persistent exposure on correct histories is necessary for the stated horizon consequence. The shared-defect example has no such exposure after its first correct output.

5 Unequal difficulty and cumulative hazard

For a deterministic locally integrable nonnegative intensity $\lambda(t)$, a nonhomogeneous Poisson process has no-fault probability $e^{-\Lambda(T)}$, where $\Lambda(T) = \int_0^T \lambda(t)\,dt$. Its raw half-success horizon is the first time Λ reaches $\log 2$, if it ever does. This extends the baseline without adding dependence. An intensity depending on random history is not interchangeable with this deterministic function.

Proposition 9 (A vanishing-segment extension of the stealth ceiling)**.** *Partition $[0, T]$ into segments with cumulative intensities $L_i = \int_{t_{i-1}}^{t_i} \lambda(t)\,dt$ and raw badness $p_i = 1 - e^{-L_i}$. Suppose segments are independent; within each segment, retries are fresh, good candidates always pass, and a bad candidate passes with probability $\sigma \in (0, 1]$. Then*

$$S(T) = \prod_i \frac{1 - p_i}{1 - p_i + p_i\sigma}. \tag{4}$$

As $\max_i L_i \to 0$, at fixed finite $\Lambda(T)$, $S(T) \to e^{-\sigma\Lambda(T)}$. Thus the limiting gated horizon is $H_\sigma = \inf\{t : \Lambda(t) \geq \log 2/\sigma\}$, with $\inf \varnothing = \infty$; when it is finite, $\Lambda(H_\sigma) = \log 2/\sigma$. A bounded Λ can give a finite raw horizon and an infinite gated one: $\lambda(t) = e^{-t}$ and $\sigma = 1/2$ give $e^{-\sigma\Lambda(t)} > e^{-1/2} > 1/2$. If $\Lambda(t) = (t/\eta)^k$, with $k, \eta > 0$, the limiting ratio is

$$H_{\text{gated}}/H_{\text{raw}} = \sigma^{-1/k}. \tag{5}$$

For $\sigma < 1$, the ratio $1/\sigma$ is recovered only at $k = 1$ in this family.

Proof. Proposition 5 gives accepted slip $p_i\sigma/(1-p_i+p_i\sigma)$. Independent segments give the displayed product. Its ith logarithm is $-L_i - \log(1 - (1 - \sigma)(1 - e^{-L_i})) = -\sigma L_i + O(L_i^2)$, uniformly as $\max_i L_i \to 0$. Since $\sum_i L_i^2 \leq (\max_i L_i)\Lambda(T)$, the error vanishes. Solve the limiting survival curve at one half. For the power cumulative intensity, $H_{\text{raw}} = \eta(\log 2)^{1/k}$ and $H_{\text{gated}} = \eta(\log 2/\sigma)^{1/k}$. □

Corollary 10 (Finite-segment bound under fresh Poisson retries)**.** *In the same independent-segment model, allow false rejection and any bad-accept probability $\alpha_i \geq \sigma_i \in [0, 1]$. Let acceptance have positive probability. Then*

$$S(T) \leq \prod_i \frac{1 - p_i}{1 - p_i + p_i\sigma_i} \leq \exp\left(-\sum_i \sigma_i L_i\right).$$

For constant intensity λ and stealth share $\sigma > 0$, this yields the exact bound $H \leq \log 2/(\lambda\sigma)$ at any fixed segment length, under log-linear interpolation between boundaries: for $T = (k + \theta)s$ with $0 \leq \theta < 1$, $S(T) = \left[\prod_{i=1}^k (1 - q_i)\right](1 - q_{k+1})^\theta$. With identical gate behavior this is $(1 - q)^{T/s}$, Part I's horizon convention. Other interpolation rules need not satisfy it: with $\lambda = 1$, $\sigma = 1/2$, and $s = 4$, linear interpolation crosses one half near $2.0746 > 2\log 2$. For the discrete horizon, round the number of segments up.

Proof. The accepted slip $p_i\alpha_i/[(1 - p_i)(1 - \beta_i) + p_i\alpha_i]$ increases with α_i and β_i. Set $\alpha_i = \sigma_i$ and $\beta_i = 0$ for its lower bound. The corresponding survival factor is $1/[1 - \sigma_i + \sigma_i e^{L_i}]$. Convexity of the exponential gives $1 - \sigma_i + \sigma_i e^{L_i} \geq e^{\sigma_i L_i}$. Multiply the factors. □

This is a reliability limit, not a cost-free policy: fixed positive gate cost makes arbitrarily short segments expensive. It models ideal visibility outside the stealth set and zero false rejection. A time-varying $\sigma(t)$ gives $\exp(-\int_0^T \lambda(t)\sigma(t)dt)$ under piecewise-continuous bounded marks and the

same refinement argument. Independent thinning of the underlying Poisson process gives that survival exactly, but thinning and the finite-segment retry protocol are different models.

The relevant average is $\int \lambda\sigma / \int \lambda$, not generally the ordinary time-average of σ. On two equal intervals, rates $(0.01, 0.09)$ and stealth shares $(0.1, 0.5)$ give mean escape intensity 0.023. Multiplying their separate means gives 0.015. High fault intensity and low visibility can coincide.

The Weibull ratio (5) is one instance of a general ordering. We use the standard reliability classes: increasing failure rate (IFR) and decreasing failure rate (DFR) [10].

Proposition 11 (Monotone hazard orders the multiplier)**.** *Let Λ be continuous and nondecreasing with $\Lambda(0) = 0$. For $\sigma \in (0, 1]$ put $H_{\text{raw}} = \inf\{t : \Lambda(t) \geq \log 2\}$ and $H_\sigma = \inf\{t : \Lambda(t) \geq \log 2/\sigma\}$. If λ is nondecreasing (IFR), then $H_\sigma \leq H_{\text{raw}}/\sigma$. If λ is nonincreasing (DFR), then $H_\sigma \geq H_{\text{raw}}/\sigma$. Under IFR and the model of Corollary 10 with $\sigma_i = \sigma$ and equal segment length s, the discrete gated horizon, measured in work, is at most $s\lceil H_\sigma/s \rceil \leq s\lceil H_{\text{raw}}/(\sigma s) \rceil$.*

Proof. Under IFR, Λ is convex with $\Lambda(0) = 0$, so $\Lambda(t)/t$ is nondecreasing for $t > 0$. If $H_{\text{raw}} < \infty$, continuity gives $\Lambda(H_{\text{raw}}) = \log 2$, and at $t = H_{\text{raw}}/\sigma \geq H_{\text{raw}}$ we get $\Lambda(t) \geq \Lambda(H_{\text{raw}})/\sigma = \log 2/\sigma$. Under DFR, $\Lambda(t)/t$ is nonincreasing. For $H_{\text{raw}} \leq t < H_{\text{raw}}/\sigma$, $\Lambda(t) \leq t \log 2/H_{\text{raw}} < \log 2/\sigma$; for $t < H_{\text{raw}}$, $\Lambda(t) < \log 2 \leq \log 2/\sigma$. For the last claim, Corollary 10 gives $S(T) \leq e^{-\sigma\Lambda(T)}$ at every segment boundary. At the first boundary $T \geq H_\sigma$, this is at most one half. The rounding is necessary: with constant rate 1, $\sigma = 1/2$, and $s = 1$, one segment survives with probability $2/(1 + e) > 1/2$, so the discrete horizon is $2 > 2 \log 2$. □

Part IV's load-monotonicity hypothesis, together with a nondecreasing load (its R-E case (a)), makes the state-carrying agent IFR. That result is ungated. If the agent is also gated under the independent-segment, fresh-retry model of Corollary 10 with constant σ, Part I's $1/\sigma$ multiplier is an upper bound for it, up to the rounding above. Bounded-load holes (R-E case (b)) bound the hazard but do not order it.

Counterexample 12 (Constant expected arrival rate does not determine a horizon)**.** Draw Z once: $Z = 0$ with probability 0.9, and $Z = 10\lambda$ otherwise, with $\lambda > 0$. Conditional on Z, arrivals are homogeneous Poisson of rate Z. Then $\mathbb{E}N(T) = \lambda T$, but $\mathbb{P}(N(T) = 0) = 0.9 + 0.1e^{-10\lambda T} > 0.9$. The half-success horizon is infinite. An ordinary rate-λ Poisson process has the same expected counts and finite horizon $\log 2/\lambda$. For any nonnegative random rate Z, $\mathbb{E}e^{-ZT} \geq e^{-\mathbb{E}[Z]T}$ by Jensen's inequality. This compares zero arrivals at fixed mean rate, not multi-fault damage or retry cost.

A latent rate changes the horizon value. If visibility does not depend on the latent rate, the rounded $1/\sigma$ ceiling survives at every segment length, and in the ideal vanishing-segment limit the multiplier is exactly $1/\sigma$. At finite segment length the exact multiplier is not invariant: with ideal gates at a deterministic rate $z > 0$, log-linear interpolation gives $H_{\text{gated}}/H_{\text{raw}} = zs/\log(1 - \sigma + \sigma e^{zs})$, which depends on z when $0 < \sigma < 1$. A numerical ratio also requires finite horizons.

Proposition 13 (Rate mixing preserves the multiplier)**.** *Draw $Z \geq 0$ once per run. Conditional on Z, let the independent-segment model of Corollary 10 hold with constant intensity $Z\lambda_0$ and a stealth share $\sigma \in (0, 1]$ that does not depend on Z. Let $S_{\text{raw}}(t) = \mathbb{E}e^{-Z\lambda_0 t}$ be the ungated survival, and let segments have equal length s. Then, at every segment boundary T,*

$$S_{\text{gated}}(T) \leq S_{\text{raw}}(\sigma T), \qquad \textit{hence} \qquad H_{\text{gated}} \leq s\,\lceil H_{\text{raw}}/(\sigma s) \rceil$$

for the discrete horizon measured in work. For ideal gates ($\alpha = \sigma$, $\beta = 0$, as in Proposition 9), the first inequality becomes an equality in the vanishing-segment limit. Without ideal gates it need not: with $\sigma = \alpha = 1/2$ and $\beta = 1/2$, each accepted-segment survival is exactly $e^{-Z\lambda_0 s}$, so $S_{\text{gated}}(T) = S_{\text{raw}}(T)$. If instead $\sigma = \sigma(Z)$, then $S_{\text{gated}}(T) \leq \mathbb{E}e^{-\sigma(Z)Z\lambda_0 T}$, and no fixed multiplier follows.

Proof. Corollary 10, applied conditional on Z, gives $S_{\text{gated}}(T \mid Z) \le e^{-\sigma Z \lambda_0 T}$. Take expectations. At the first boundary $T \ge H_{\text{raw}}/\sigma$, $S_{\text{raw}}(\sigma T) \le S_{\text{raw}}(H_{\text{raw}}) = 1/2$ by continuity and monotonicity of S_{raw}. For the limit, apply Proposition 9 conditional on Z and use bounded convergence. An interpolated bound needs log-linear interpolation of the conditional boundary survivals before averaging. □

Without rounding the horizon bound is false at fixed s: the example in the proof of Proposition 11 is the case $Z \equiv 1$. The survival form also covers Counterexample 12, where both horizons are infinite. With $Z \in \{0.5, 1.5\}$ equally likely, $\lambda_0 = 1$, and ideal gates, independent visibility $\sigma = 0.2$ gives multiplier 5.0000 in the limit. Visibility $\sigma(Z) = (0.1, 0.3)$, with the same mean 0.2, gives 5.0549. Together with Proposition 11, for the ideal vanishing-segment survival curve: random rate mixing keeps the multiplier, deterministic IFR lowers it, DFR raises it, and visibility that co-varies with difficulty removes any fixed value. At finite segment length the gated horizon can be shorter: with constant rate 1 (both IFR and DFR), ideal gates, $\sigma = 1/2$, and $s = 1.2$, one segment survives with probability $2/(1+e^{1.2}) < 1/2$, so the discrete horizon is $1.2 < 2 \log 2$. The DFR lower bound concerns H_σ, not finite-segment horizons.

6 Checkpoint cost and persistent state

Proposition 14 (Heterogeneity changes mean retry cost)**.** *Consider equal-length segments $s > 0$, perfect detection, no false rejection, independent fresh retries within each segment, and a segment-specific rate $L \ge 0$ that stays fixed during its retries. Each attempt costs $s + G$. If the workload distribution of rates has finite mean, then*

$$C(s) = (1 + G/s)\mathbb{E}[e^{Ls}] \ge (1 + G/s)e^{\mathbb{E}[L]s}.$$

Equality holds only for a constant rate (when $s > 0$); the left side can be infinite. If the rate distribution does not depend on s, and the moment and its derivative are finite near an interior optimizer, the stationarity condition is

$$\frac{\mathbb{E}[Le^{Ls}]}{\mathbb{E}[e^{Ls}]} = \frac{G}{s(s+G)}. \tag{6}$$

Proof. A rate-L attempt passes with probability e^{-Ls}. Expected attempts are e^{Ls}. Average the conditional cost and apply strict convexity of the exponential. Differentiating $\log C(s)$ gives the stationarity equation under the stated moment conditions. □

For constant $L = \lambda$, the exact stationary equation in this model is $s(s+G) = G/\lambda$. Its positive root approaches $\sqrt{G/\lambda}$ when $G\lambda \to 0$. This is the model used for Part I's approximation. Other detection times and recovery costs give other checkpoint formulas; general failure distributions already have a large checkpointing literature [8].

At $s = 1, G = 0.1$, equal workload shares with rates 0 and 2 give $C = 1.1(1 + e^2)/2 \approx 4.6140$. Replacing them by their mean rate 1 gives $1.1e \approx 2.9901$. The inequality of Proposition 14 holds at every s, so it also holds between the two optimal costs. With the two rates held fixed as s varies, the optimal gap is much smaller: about 1.8530 at $s \approx 0.239$ against 1.7951 at $s \approx 0.270$, because the heterogeneous optimum uses shorter segments. The fixed-interval figure alone overstates the practical effect. No randomness is needed. Take a deterministic intensity $\lambda(t)$ indexed by work position, and let a retry repeat the same position, so that it keeps the same rate. A segment's rate is then the average of λ over that segment. At each fixed s these averages have mean equal

to the overall mean rate, so the same Jensen comparison holds at every s. Their distribution changes with s, however, so Equation (6) does not apply directly; the derivative must include that dependence. Part I's limitations paragraph states that a clustered or time-varying hazard leaves the mean checkpointing overhead governed by the same λ and inflates only its variance. The countermodel above assumes that a retry keeps its segment's rate, as a retry of the same hard step would. The paragraph does not state that assumption, but its own mechanism, runs of bad segments that trigger consecutive recomputations, is consistent with it. Under that assumption the statement is false and the mean rises. This example changes no gate behavior and uses fresh retries within each segment. If the rate is redrawn independently on each attempt instead, perfect detection gives expected attempts $1/\mathbb{E}[e^{-Ls}]$. The reset law therefore changes even the direction of a comparison with mean-rate substitution.

Finite state models. A latent-state model can include context poisoning, repair, and reset without pretending that state is an average rate. Let X_i be a finite state before emitted unit i and let

$$K_i(x, y) = \mathbb{P}(I_i = 0, X_{i+1} = y \mid X_i = x)$$

under a Markov assumption that makes X_i sufficient for this transition. K_i is substochastic. For initial row distribution π,

$$S_n = \pi K_1 K_2 \cdots K_n \mathbf{1}. \tag{7}$$

This follows by summing probabilities of correct state paths. For retries within one unit, let $R(x, y)$ be the joint probability of rejection and next state y, and let $b(x)$ be immediate bad acceptance. If the spectral radius of R is below one, the eventual bad probability vector is $(I-R)^{-1}b$, and expected attempts are $(I - R)^{-1}\mathbf{1}$. Both follow by summing the convergent series $\sum_{r\geq 0} R^r$. Perfect reset, retained premises, and learning correspond to different kernels. These formulas do not establish which kernel is appropriate for an agent.

Propagation. One escaped root fault can corrupt many descendants. A model of fresh introduction counts the root once; a model of faulty outputs counts its consequences. For the metric of Definition 1, what happens after the first escaped fault cannot change the already-failed prefix. It can change damage and repair cost. Excitation by a detected or otherwise unescaped fault is different: it acts before any escape and changes the conditional prefix risks. If a root fault occurs with probability p and escapes with probability σ, and a detected root causes one further faulty attempt that escapes with probability σ, the escape probability is $p\sigma + p(1 - \sigma)\sigma$, not $p\sigma$. A self-exciting arrival model can represent some forms of increased subsequent fault activity [9], but does not by itself identify distinct causes, visibility, or semantic propagation.

7 Mechanization with heterogeneous risks

At a correct pre-unit history h, let $c_i(h)$ be the probability of routing to a mechanism. Let $o_i(h)$ be the open-path slip conditional on choosing that path. On the mechanism branch, let $e_i(h)$ be direct escaped-fault probability and $b_i(h)$ the probability of rejection into fallback. Let $f_i(h)$ be the fallback slip conditional on that rejection. Direct escape and rejection are disjoint events.

Proposition 15 (Routing identity and a restricted closure comparison)**.** *At each such history, the slip probability is*

$$q_i(h) = (1 - c_i(h))o_i(h) + c_i(h)\{e_i(h) + b_i(h)f_i(h)\}. \tag{8}$$

Averaging over histories conditional on A_{i-1} gives the q_i of Proposition 2. For a fixed exogenous sequence of independent units, suppose routing changes neither later input laws nor branch risks. Put $d_i = e_i + b_i f_i$. Replacing the open path at unit i by the mechanism improves its survival factor exactly when $d_i < o_i$. It strictly improves total finite-run survival if the other factors are positive. For stationary identical units this recovers Part IV's realized-slip criterion and product law.

Proof. Condition first on routing, then on rejection within the mechanism branch. This gives the identity. Under the additional comparison assumptions, total survival is the product of fixed local factors. Replacing $1 - o_i$ by $1 - d_i$ has the stated effect. □

The comparison assumptions matter. A mechanism with lower immediate risk can move execution to a state with higher later risk. For example, an open first step can have risk 0.1 and leave a safe second step. A closed first step can have risk zero but leave a second-step risk 0.9. Total survival falls from 0.9 to 0.1. Equation (8) remains true; the claim that local improvement implies global improvement does not.

Work share and risk share. For small risks on a fixed independent workload, $1 - S_n = \sum_i q_i + O((\sum_i q_i)^2)$. For units with small p_i, fresh retries, ideal acceptance outside a common stealth set ($\alpha_i = \sigma$, $\beta_i = 0$), and closed units with zero escape, the first-order remaining risk is $\sigma \sum_i (1 - c_i) p_i$. A stealth floor alone gives only a lower-bound contribution: with $\sigma = 0.1$ but bad acceptance 0.5, the open slip is $0.5 p_i + O(p_i^2)$. Define the fault-exposure share $c_{\text{risk}} = \sum_i c_i p_i / \sum_i p_i$ when the denominator is positive. It equals work share only under additional homogeneity or selection assumptions. Consider one unit with risk 0.01 and one with risk 0.09. Closing either one gives 50% work closure, but leaves risk 0.09 or 0.01. The original PSM formula is appropriate for its stationary mixing model; volume share alone does not generalize it to this workload. These risks describe escapes, not confidence scores.

Costs under changing demand. For deterministic exogenous call intensity $r(t)$, expected per-call saving $\Delta(t)$, upkeep rate $u(t)$, fixed lifetime τ, build cost B, and validation cost V, linear accounting gives closure benefit

$$\int_0^\tau [r(t)\Delta(t) - u(t)]dt - B - V. \tag{9}$$

Here per-call saving includes the actual expected retry, fallback, and loss costs on each path. With random lifetime and demand, use the expectation of the stopped total; products of means need not agree. Part IV's $(r\Delta - u)\tau > B + V$ is the constant-parameter special case. Selecting regions separately remains valid only with additive costs and no coupling.

8 Blind sets, evidence, validation, and drift

Proposition 16 (A structural blind set under changing fault laws)**.** *Let Σ be a measurable fault set accepted by every gate in a fixed allowed collection. For every fault distribution μ_t and every stack from that collection, its conditional false-accept mass is at least $\mu_t(\Sigma)$. For two ideal stacks accepting exactly fault sets Σ_F, Σ_N, the combined mass is $\mu_t(\Sigma_F \cap \Sigma_N)$. It equals the product of the two masses only under independence of these events under that same μ_t.*

Proof. The combined accepted-fault set contains Σ. Apply monotonicity of measure. For the two ideal stacks, conjunction is set intersection; independence is precisely the condition giving the product of masses. □

Part II supplies the inclusion from its representation and zero-false-rejection assumptions. Its finite false-rejection extension also needs the stated likelihood-ratio control. Arrival stationarity is not used. A constant numerical floor does require a lower bound on $\mu_t(\Sigma)$ in the relevant conditional workloads. A changed representation, predicate, or allowed evidence class can change Σ itself.

Proposition 17 (Twins under dependent observation)**.** *Fix a target distribution and a mechanism class with fault mass $\pi(M)$. Suppose two mechanisms produce the same law for the full observation record and auxiliary randomness used by a promotion rule. Their promotion probabilities are equal. Let $U = u(\text{record}, \text{auxiliary randomness})$ for one fixed measurable rule u. If this upper bound satisfies $\mathbb{P}_M\{\pi(M) \le U\} \ge 1 - \eta$ for every mechanism in the class, then*

$$\mathbb{P}_M\Big\{U \ge \sup_{M' \sim M} \pi(M')\Big\} \ge 1 - \eta,$$

where $M' \sim M$ denotes equality of those observation laws. Pointwise twins have equal observation laws under any common exogenous sampling law, even with dependent samples, provided the joint auxiliary-randomness law is also common.

Proof. The same measurable rule applied to identically distributed records has identical laws. The common rule is necessary: an oracle bound $U = \pi(M)$ is valid for every mechanism but is not a function of the record, and it would separate twins of masses 0 and 1. For each t below the supremum, choose a twin with fault mass greater than t and apply validity for that twin. Transfer the distribution of U to M. The events $\{U \ge t\}$ decrease as t increases to the supremum along a sequence, and their intersection is $\{U \ge \sup_{M' \sim M} \pi(M')\}$. Continuity of measure from above gives the displayed bound. Pointwise twins agree on every realized sample, so common joint sampling and randomness give the required equality in law. □

Common adaptive sampling based only on the shared transcript and common independent random seeds also preserves pointwise twins, by induction on the transcript. Side information about mechanism text or identity can break this condition. Neither the result nor Part IV's original twin result forces the mechanism itself to be wrong. They constrain what the specified evidence can certify. This extension does not validate the separate generalizing-builder hypothesis.

Proposition 18 (Validation needs conditional exposure)**.** *Let D_j mean that validation item j detects a fault, with exact evidence on that item. If every candidate above target fault mass $\theta \in (0, 1)$ has*

$$\mathbb{P}(D_j \mid D_1^c, \ldots, D_{j-1}^c) \ge \theta$$

whenever the conditioning event has positive probability, then a zero-detection promotion rule falsely promotes such a candidate with probability at most $(1 - \theta)^n$. Thus $n \ge \log(1/\eta)/[-\log(1 - \theta)]$ is sufficient for error at most η.

Proof. Apply the chain rule to the event of no detections. □

Independent target-distribution samples satisfy this condition. Repeating one randomly drawn input n times does not: for a deterministic mechanism of fault mass θ' the chance of no detection is $1 - \theta'$, for every $n \ge 1$. At $\theta' = 0.1, n = 20$, the two probabilities are 0.9 and $0.9^{20} \approx 0.1216$. Part IV's sample-size necessity follows within its original class with masses arbitrarily close to θ and independent target samples. Necessity is not asserted for arbitrary adaptive or targeted validation schemes.

Proposition 19 (Distribution drift and changing fault sets)**.** *For probability measures D_t, D_0, use $d_{\mathrm{TV}}(D_t, D_0) = \sup_A |D_t(A) - D_0(A)|$. If a mechanism's escaped-fault set B is fixed, then $|D_t(B) - D_0(B)| \le d_{\mathrm{TV}}(D_t, D_0)$. If the set also changes from B_0 to B_t, then*

$$|D_t(B_t) - D_0(B_0)| \le d_{\mathrm{TV}}(D_t, D_0) + D_0(B_t \triangle B_0).$$

Proof. The first statement is the definition of total variation. For the second, add and subtract $D_0(B_t)$; the difference between masses of two sets is bounded by their symmetric difference. □

The bound needs information about drift; it does not estimate it. With unrestricted shift, a mechanism correct on all validation inputs can fail on all future inputs. A watch/reopen policy therefore needs a measurement model and delay analysis, which remain open.

9 When the earlier results hold

The companion assessment (`fault_process_assessment.json`) identifies every formal result by its source label and hash. The following tables summarize the logical dependencies. "Preserved" means preserved under the listed assumptions, not established empirically. "Replace" means the formula changes outside the original model. No published source is edited by this assessment.

Table 2 classifies the 25 results. *Arrival-free* results use no arrival or stationarity assumption in their stated conclusion; their masses still depend on the current fault law. *Conditional* results keep the form of their conclusion when uniform or conditional bounds hold on the reachable histories. *Formula* results are quantitative formulas derived under constant or stationary parameters (a constant rate, or stationary Bernoulli risks); their form changes outside the original model. The classes are applied in this order of precedence. The counts are 14, 10, and 1. A bullet marks an axis of Section 2 whose change can break the result's stated conclusion: D difficulty, P dependence, R retry, V visibility, I intervention, W drift. Bullets concern the conclusion itself, not surrounding applications; for example, Part III's cost-separation comparison is per unit, while translating an output target into a per-unit target depends on output dependence. The R bullets on the horizon ceiling and stealth decomposition concern their exact H_{raw}/σ forms; their first-attempt bounds survive any retry law. R-A is conditional because its displayed form holds whenever $q_i \ge p\lambda_{\mathrm{mix}}$ on correct prefixes (Proposition 2). The zero-escape and reach results are conditional for the same reason: a constant correct-prefix risk keeps their conclusions, even with dependent outputs. A geometric first-failure index K with $I_i = \mathbf{1}\{i \ge K\}$ gives dependent outputs and exactly $S_n = (1-f)^n$. The classification is a review judgment recorded with the assessment; it is not a mechanical proof classification.

Result	Assessment
Part I: gated computation	
Per-segment slip	Preserved with constant conditional retry probabilities. For nonfresh retries, replace $p\alpha_m/P_{\mathrm{adv}}$ by Proposition 5. Counterexample 6.
Threshold and overhead	Preserved under uniform conditional gate accuracy and progress: Theorem 7. Mean accuracy alone is insufficient, and unbounded difficulty ($p_i \to 1$) removes the uniform acceptance bound. The converse needs persistent badness and conditional escape, rather than a marginal nonzero rate.

Result	Assessment
Horizon ceiling	Replace a fixed $p\sigma$ by conditional exposure bounds (Proposition 8). A positive marginal floor alone does not imply a finite horizon. The printed first-attempt bound $s \log 2/(p\sigma)$ survives arbitrary retries; the exact H_{raw}/σ form needs fresh retries (Corollary 10) or an equally strong conditional survival bound. The $1/\sigma$ multiplier survives rate mixing (Proposition 13) and is an upper bound under IFR (Proposition 11); under DFR the multiplier can exceed it.
Diversity product	Preserved under independence of stealth events under the operative fault law. Family names or temporal independence do not establish this independence.
Stealth decomposition	Its mixture identity survives pointwise under the stated conditional family model. The appended horizon consequence needs persistent exposure. Under fresh Poisson retries the exact constant-rate bound follows from Corollary 10; a changed retry law can break it. Its breadth limit needs $\rho < 1$ (errata candidate 4).
Optimal interval	The constant-rate approximation survives in its cost model. Heterogeneity, including a deterministic $\lambda(t)$, raises mean cost at every interval and requires Equation (6) when the rate distribution does not depend on s; persistent state requires a retry model. The printed minimum drops a $1/(1-\beta_m)$ factor (errata candidate 3).
Part II: generated gates	
GG1 inheritance	Structural inclusion survives without an arrival model, under fixed representation and soundness/support assumptions. Mass is evaluated under the new fault law.
GG2 fixed point	Lower bound survives for sound stacks. Equality additionally requires reachable coverage outside the blind set. This is an existing achievability qualification, independent of arrival stationarity.
Exogenous family	Intersection identity survives for the ideal stacks. Product requires independence under the current law; strict reduction requires positive removed mass, not set noncontainment (errata candidate 5).
Reality grounding	Zero covered-scope escape survives if the oracle matches the current correctness predicate. Exact residual equality off scope requires further conditions (errata candidate 6).
Part III: escape, cost, correlation	
Series-system identity	For independent unequal marginals use $1-\prod_i(1-f_i)$. For arbitrary dependence use conditional risks or Equation (2).
Zero-escape corollary	Preserved when the correct-prefix risk is constant, $q_i = f$; independence across outputs is not needed. For varying sequential risks, summability governs positive infinite survival; a triangular array has vanishing total escape iff its conditional-risk sum tends to zero.
Cost separation	Preserved if the per-unit power law, geometric gate improvement, fixed prevalence, good-acceptance rate, and cost assumptions remain. No Poisson law is needed. The translation from output risk to per-unit target must be revised under dependence.
Association lemma	Preserved for same-direction monotone functions of independent primitives; no constant arrival rate is needed [7].
Correlated-escape bound	Preserved under that provenance model at fixed marginals. Arbitrary dependence can violate it. The bound concerns any escape, not simultaneous damage.
Part IV: progressive semantic mechanization	

Result	Assessment
R-A partial closure	Original ceiling needs independent stationary mixing and fixed mechanisms. Use conditional routing risks for history dependence. Work share alone is insufficient. Its $(1+O(\lambda s))$ factor is necessary for the closed branch (errata candidate 2).
Evidence twins	Preserved under equal observation laws. Independent samples can be weakened as in Proposition 17; changes that distinguish twins are outside that premise.
Same-family evidence	Preserved under the representation restriction; the explicit twin-mass formula also needs the countable scope and finite-patch closure. Drift changes its target mass. Evidence not decided within F, such as exact correctness evidence, can separate F-twins.
Exact replay	The disagreement bound is pointwise and survives every fixed target law. It must be recomputed or bounded under shift.
Accepted-output copy	The displayed lower bound $p\lambda_F - \mathbb{E}[\delta]$ survives without fresh retries: under eventual emission a first-attempt stealth fault is emitted at once. Fresh retries only give the proof's intermediate denominator step. An exact reference gate removes the floor. No realized-build floor follows.
Produced undetected faults	Preserved with the build/input law, preservation of undetected outputs, and no discarded builds. Repair that changes undetected outputs can break it. Both parts depend on visibility: detecting previously invisible build faults removes the floor.
R-C break-even	Accounting survives with actual expected costs; use Equation (9) for varying demand. Conditional exposure gives a sufficient zero-failure sample count (Proposition 18); the source's necessary count needs its independent target sampling and near-threshold class, and targeted validation can need far fewer items. Separable frontier selection also needs additive uncoupled costs.
R-D composition	Preserved under residual completeness and monotone provenance. Without provenance association, completeness still gives a union bound. Missing interaction faults can invalidate completeness itself.
Reach and residual	Exact geometric horizon and $1/(1-c)$ comparison need the stated fixed branch parameters and a constant correct-prefix risk $q(c)$; independence across outputs is not needed. Feedback and selected routing require a new comparison.
R-E bounded load	Already uses a varying load-dependent Poisson rate. Its bounds survive with the stated load/intensity bounds and clock. Bounded load alone establishes neither monotonicity nor clean reset. With nondecreasing load the ungated agent is IFR; if it is also gated under the model of Corollary 10, Proposition 11 makes Part I's $1/\sigma$ an upper bound on the multiplier, up to rounding.

Errata candidates in the published papers. The assessment found seven items in Parts I and II. Each carries a severity: a *false claim* within the stated model, a *strengthening* of a correct bound, a *dropped factor*, an *overstated equality or strictness*, or an *endpoint* restriction. Only item 1 depends on the arrival model. These qualifications concern the cited versions; this paper does not amend them.

1. *Part I, limitations: mean checkpoint overhead (false claim, prose).* The claim that a clustered or time-varying hazard changes only the variance of checkpointing overhead is false when a retry keeps its segment's rate, including for a deterministic $\lambda(t)$ (Proposition 14 and Section 6). The paragraph does not state that retry assumption; the countermodel adds it. The effect at the respective optimal intervals can be small.

Table 2: Classification of the earlier results and the axes that can break them.

Part	Result	Class	D	P	R	V	I	W
I	Per-segment slip	conditional			•			
I	Threshold and overhead	conditional	•	•	•	•		
I	Horizon ceiling	conditional	•	•	•	•		
I	Diversity product	arrival-free		•		•		
I	Stealth decomposition	conditional	•	•	•	•		
I	Optimal interval	formula	•		•			
II	GG1 inheritance	arrival-free				•		•
II	GG2 fixed point	arrival-free				•		•
II	Exogenous family	arrival-free		•		•		
II	Reality grounding	arrival-free				•		•
III	Series-system identity	conditional	•	•				
III	Zero-escape corollary	conditional	•	•				
III	Cost separation	arrival-free						
III	Association lemma	arrival-free		•				
III	Correlated-escape bound	arrival-free		•				
IV	R-A partial closure	conditional	•	•		•	•	•
IV	Evidence twins	arrival-free						•
IV	Same-family evidence	arrival-free				•		•
IV	Exact replay	arrival-free						•
IV	Accepted-output copy	arrival-free				•		
IV	Produced undetected faults	arrival-free				•	•	
IV	R-C break-even	conditional		•			•	•
IV	R-D composition	arrival-free		•				
IV	Reach and residual	conditional	•	•			•	
IV	R-E bounded load	conditional	•				•	

2. *Part I, horizon ceiling: the $(1 + O(\lambda s))$ factor (strengthening).* The theorem prints $H \le s \log 2/(p\sigma) = (\log 2/(\lambda\sigma))(1 + O(\lambda s))$, from the floor $S_n \le (1 - p\sigma)^n$. That bound is correct. Retaining the acceptance denominator gives Corollary 10: under Part I's fresh-retry assumptions and its continuous horizon convention, $H \le H_{\text{raw}}/\sigma$ holds at every segment length, so the factor can be removed. The discrete horizon rounds up to whole segments. The stealth decomposition prints the exact form $H_{\text{gated}} \le H_{\text{raw}}/\lambda_{\text{irr}}$, but its proof substitutes into the first-order ceiling; Corollary 10 supplies the exact form under fresh retries. Without fresh retries the exact form can fail while the printed first-order bound holds: with $\lambda = s = 1$, $\sigma = 1/2$, stealth faults accepted, and a rejected nonstealth fault followed by a good retry, $H \approx 1.8246 > H_{\text{raw}}/\sigma \approx 1.3863$, while $s \log 2/(p\sigma) \approx 2.1931$. Part IV's closed paths cannot inherit this. With $\varepsilon_{\text{cl}} = p\lambda_{\text{pc}}$ and $0 < \lambda_{\text{pc}} \le 1$, the function $t \mapsto -\log(1 - t(1 - e^{-L})) - tL$ is convex on $[0, 1]$ and vanishes at both ends. Hence $-\log(1 - p\lambda_{\text{pc}}) \le \lambda_{\text{pc}} L$, and an all-closed run has exact horizon at least $H_{\text{raw}}/\lambda_{\text{pc}}$. At $\lambda s = 1$ and $\lambda_{\text{pc}} = 1/2$ it is $1.3162\, H_{\text{raw}}/\lambda_{\text{pc}}$. The $(1 + O(\lambda s))$ factor of R-A is therefore necessary for its closed component; the open component can use Corollary 10.

3. *Part I, optimal interval: the minimal overhead (dropped factor).* The printed cost is $C(s) \approx (1 + G/s)e^{\lambda s}/(1 - \beta_m)$, but the printed minimum $1 + 2\sqrt{G\lambda}$ omits the factor $1/(1 - \beta_m)$. For fixed $\beta_m = 1/4$ the minimum tends to $4/3$, not 1, as $G\lambda \to 0$. The corrected statement is $C_{\min} = (1 + 2\sqrt{G\lambda} + O(G\lambda))/(1 - \beta_m)$, or the theorem should assume small β_m. The minimizing interval is unaffected.

4. *Part I, stealth decomposition: convergence (endpoint).* The limit $\lambda_{\text{irr}} + (1 - \lambda_{\text{irr}})\rho^k \to \lambda_{\text{irr}}$ needs

$\rho < 1$ (or $\lambda_{\text{irr}} = 1$); at $\rho = 1$ breadth does not help.

5. *Part II, exogenous family: strict reduction (overstated strictness).* Set noncontainment does not imply a strict reduction in mass. Under uniform measure on $[0, 1]$, $\Sigma_F = [0, 1/2]$ and $\Sigma_N = [0, 1/2)$ satisfy $\Sigma_F \not\subseteq \Sigma_N$, but both masses are $1/2$. The condition should be $\mu(\Sigma_F \setminus \Sigma_N) > 0$. Part II's model section states that equality between fault sets may be read up to μ-null sets; extending that reading to containment also repairs the statement, so this item is a wording repair.

6. *Part II, grounding proposition: off-scope residual (overstated equality).* The proposition guarantees that no fault passes on the covered scope. An oracle declared arbitrary off scope can reject additional stealth faults there, so the displayed off-scope residual mass is not always an equality. Conversely, a nonideal family stack can pass additional nonstealth faults off scope. If the family stack accepts exactly Σ_F and the oracle accepts everything outside S, residual escape is exactly $\mu(\Sigma_F \setminus S)$. Under the first condition alone it is at most that mass. The covered-scope guarantee is unaffected.

7. *Part II, GG2: equality needs reachable coverage (overstated equality; already flagged).* The equality needs sufficient reachable gates to reject faults outside the stealth set. The published repository already flags this. Its lower-bound proof uses sound gates. Separate per-gate false-rejection tolerances do not automatically give a fixed tolerance for an arbitrarily large conjunction.

What has not been proved. We do not prove that real faults follow any listed process, that resets remove hidden state, that real gate stacks satisfy a uniform conditional edge, or that drift is observable in time. Part IV's generalizing-builder and load-monotonicity hypotheses remain hypotheses. An arbitrary adaptive provenance graph need not satisfy the monotone primitive model. The evidence ladder and typed-evidence framing in Part III are not probability theorems and do not become invalid merely because an arrival model changes.

10 Executable checks and counterexamples

The companion script (`simulate_fault_process.py`) enumerates small probability spaces, checks closed forms against direct enumeration, and evaluates the examples below. Checks are deterministic. They do not establish the universal propositions; the proofs do that subject to their stated assumptions. Negative controls deliberately reinstate false independence, pooled-retry, sample-count, or fixed-multiplier claims and must be rejected by the check runner.

Comparison	First model	Second model
100 outputs, marginal risk 0.01: independent / shared survival	0.3660	0.9900
Persistent retry example: pooled prediction / actual bad fraction	0.3333	0.5000
Same example: pooled prediction / actual expected attempts	1.3333	1.5000
Mean rate / unequal rates: overhead, $s = 1, G = 0.1$	2.9901	4.6140
20 validation items, risk 0.1: independent / repeated no-detection	0.1216	0.9000
Close half of two units: close low-risk / close high-risk, remaining risk	0.0900	0.0100
Optimal overhead, $G = 0.1$: mean rate / unequal rates	1.7951	1.8530
Rate mixing, $\sigma = 0.2$: multiplier, independent / Z-dependent visibility	5.0000	5.0549
Weibull, $\sigma = 0.2$: multiplier, IFR $k = 2$ / DFR $k = 1/2$	2.2361	25.0000
Closed path, $\lambda s = 1$, $\lambda_{\mathrm{pc}} = 1/2$: horizon in units of $H_{\mathrm{raw}}/\lambda_{\mathrm{pc}}$, first-order / exact	1.0000	1.3162

The assessment manifest binds each prior-result row to the exact source snapshot. A coverage check rejects a missing formal result or a changed source hash. This is a completeness and provenance check, not an automated judgment of the source proofs. The local verification receipt records check output, negative-control exits, and source hashes. No empirical dataset is introduced.

11 Related work and limitations

The constant-hazard baseline is motivated by Ord's task-duration analysis [5]. Hoeffding supplies the exponential majority-error bound [6]. Association and the fixed-marginal reliability bound are classical [7]. Checkpoint placement for nonidentical tasks and general failure distributions predates this series [8]. Hawkes processes provide one established family for history-dependent arrivals [9]. The IFR and DFR classes are standard in reliability theory [10]. Frailty models show that mixing individual hazards changes the population hazard [11]; Proposition 13 is the corresponding statement for the gating multiplier. Nonhomogeneous Poisson models of software failures [12], imperfect repair [13], and Markov-modulated Poisson processes [14] supply the other candidate families of Table 1. None is introduced here as a new stochastic process.

Our contribution is an assumption-separated application to the interaction of imperfect gates, retries, representation floors, and mechanization. The mathematical results are elementary. The literature check is scoped, not a systematic novelty review. This is a preprint; it has not been peer reviewed.

The analysis uses an all-correct prefix metric, eventual emission except in explicit counterexamples, and declared clocks. It does not infer an arrival process from benchmark success rates. It does not optimize a general adaptive policy, price correlated catastrophic loss, or analyze a watch policy's detection delay. Numerical parameters are illustrative. Changing a model outside its assumptions does not overturn a result within those assumptions.

12 Conclusion

Constant fault intensity is not the common foundation of every result in the series. Of the 25 named results, 14 are arrival-free, 10 survive under stated conditional bounds, and 1 is a stationary-parameter formula that needs replacement outside its original model. Conditional accuracy and progress suffice for a logarithmic verification threshold. For the ideal gated survival curve, the $1/\sigma$ ceiling multiplier survives random rate mixing, is an upper bound under an increasing failure rate, and can be exceeded under a decreasing one; at finite segment length the upper bounds hold after rounding. Persistent conditional exposure, rather than a marginal floor alone, creates a horizon ceiling. Blind-set and evidence-indistinguishability results survive many changes in arrivals, while their numerical masses depend on the workload. Mechanization changes both exposure and dependence. Its effect requires branch risks, fallback, state transitions, and maintenance costs to be stated. These distinctions specify when Parts I–IV apply and where a different model is required.

Code and data availability

The standard-library Python script `simulate_fault_process.py` and the assessment `fault_process_assessment.json` reproduce the checks and tables of Section 10 and the result classification of Section 9. Both are at https://github.com/ivmat/gated-computation-sim. This paper is archived on Zenodo, DOI 10.5281/zenodo.23293740.

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